

2. IoT Architecture

Internet of Things

Executive Summary

2

Motivation: IoT emerges as a paradigm to connect physical devices (“things”) with computing and communication capabilities, enabling smart systems and services.

Problem: It is necessary to define IoT architectures, understand computing paradigms (edge, fog, cloud), and relate them to practical implementations.

Overview:

- Introduction to layered IoT architectures.
- Discussion of standards such as oneM2M.
- Comparison of edge, fog, and cloud computing paradigms.
- Hands-on lab: diode operation and rectification.

Conclusion: IoT integrates architectural design, distributed computing, and hardware fundamentals to enable efficient and scalable smart systems.

Outline

3

Introduction

IoT Architecture

Initiatives for IoT Architectures

Edge Computing

Lab: Diode

Conclusions

Recall: IoT Architecture (as a service)

4

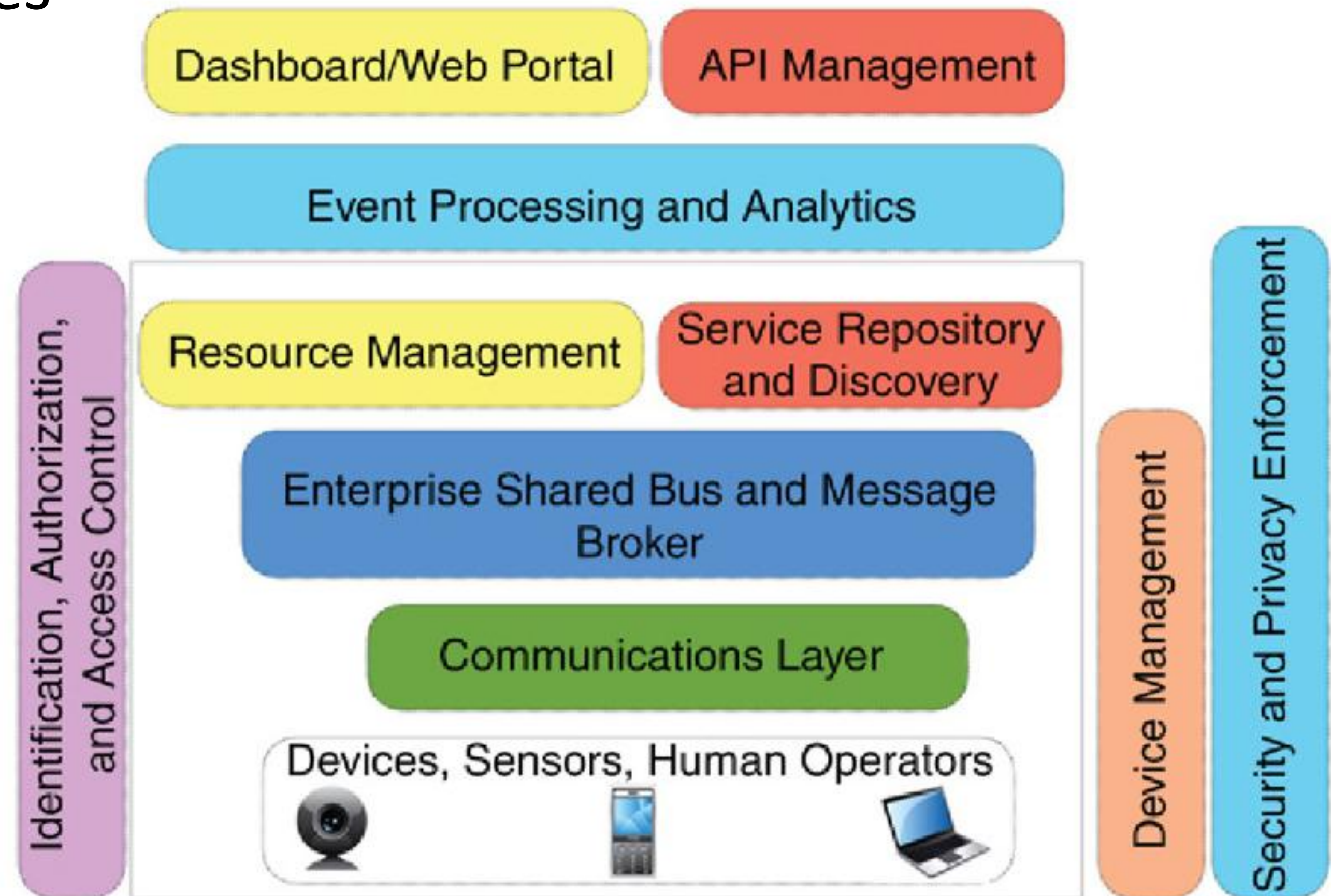
. Service Oriented Architecture (SOA):

1. **Sensing layer** is integrated with available hardware objects to sense the status of things
2. **Network layer** is the infrastructure to support over wireless or wired connections among things
3. **Service layer** is to create and manage services required by users or applications
4. **Interfaces layer** consists of the interaction methods with users or applications

Recall:Reference IoT implementation

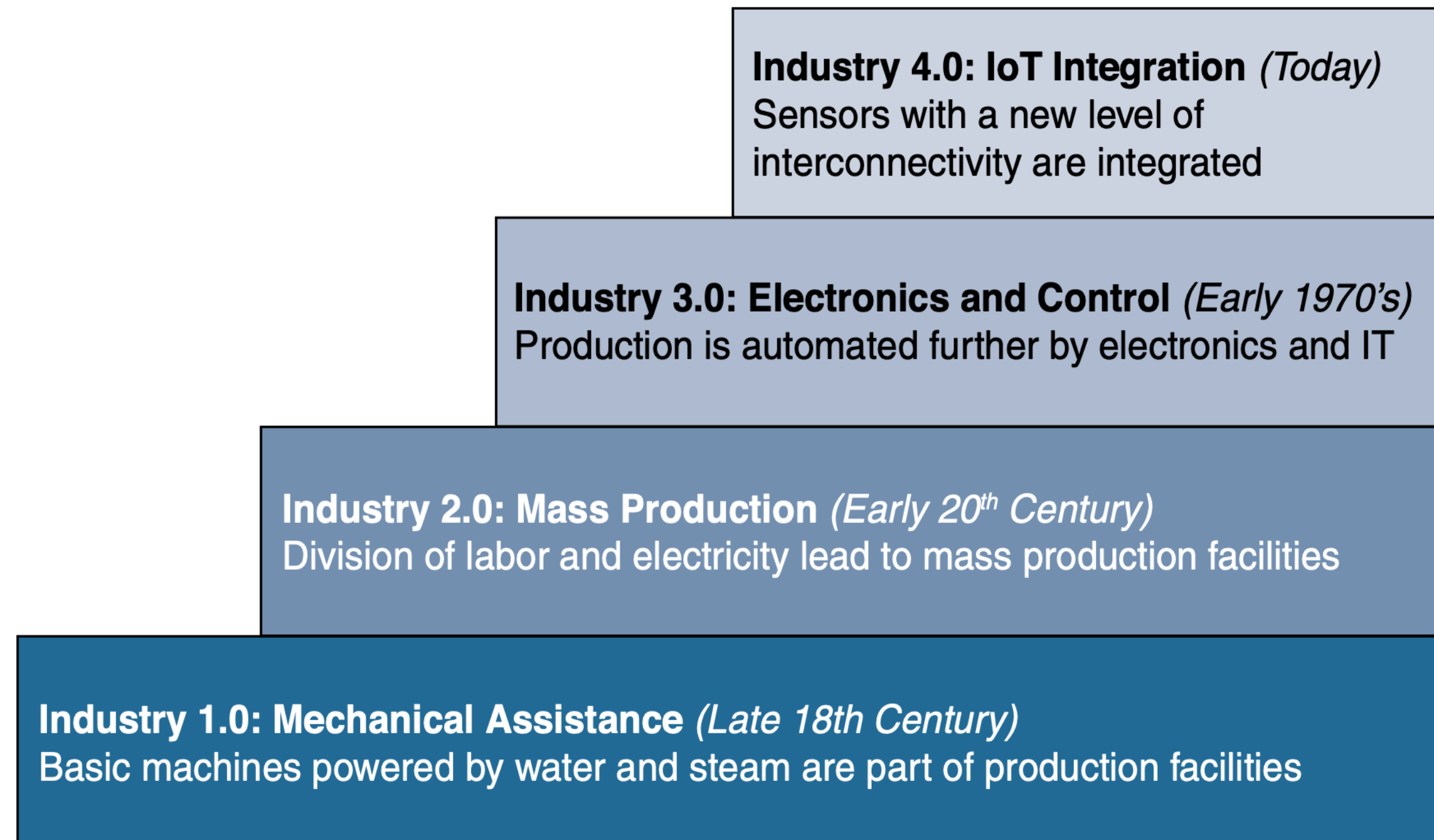
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- Executable program that complies with the I/O behavior of the standard.
 - May be written in a variety of language.
- Standards for communication, sensing, privacy, etc.



IoT Integration

6



General requirement:

1. Machine to Machine (M2M) Communication
2. Big data analysis

Three domains:

1. monitoring and actuating
2. business process and data analysis
3. information gathering and collaborative consumption.

Specific applications :

- logistics and transportation, supply-chain management, fleet management
- enterprise automation systems (smart farming, aviation industry)
- healthcare systems
- smart cities and buildings, social IoT
- disaster management, environmental monitoring, smart watering, etc.

Outline

7

Introduction

IoT Architecture

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Traditional IT vs. IoT Architecture

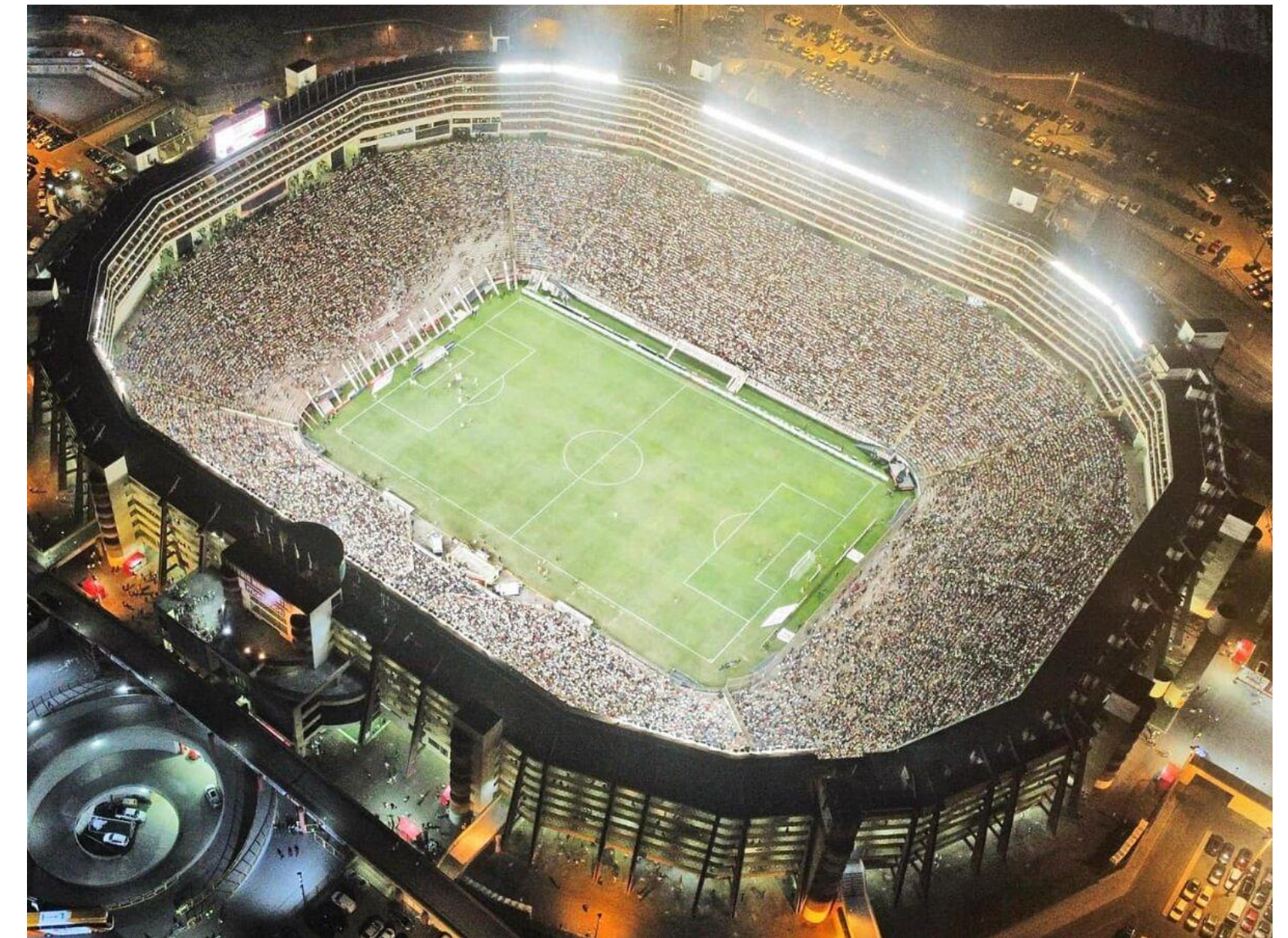
8

Traditional IT → Designing a House

- Built for a **limited number of users and devices**
- Controlled environment
- Security and functionality are easier to manage

IoT → Designing a Massive Stadium

- **Thousands → millions of endpoints**
- Highly diverse devices and usage patterns
- Much greater **scale, exposure, and complexity**



IoT is not simply “more IT devices” — it changes the architectural problem.

It's All About the Data

Traditional IT

- . Supports **business applications**
- . Email, web, CRM, databases
- . Focus: **reliable and continuous service**

IoT

- . Starts with **data generated by sensors**
- . Data must be:
Transported → Collected → Analyzed → Acted Upon

Move data from the physical world to meaningful action.

The Four Drivers of IoT Architecture

10

1. MASSIVE SCALE

Thousands → Millions of endpoints
IPv6 and scalable addressing

2. SECURITY

Exposed and physically accessible devices
Larger attack surface and new threats

3. CONSTRAINTS

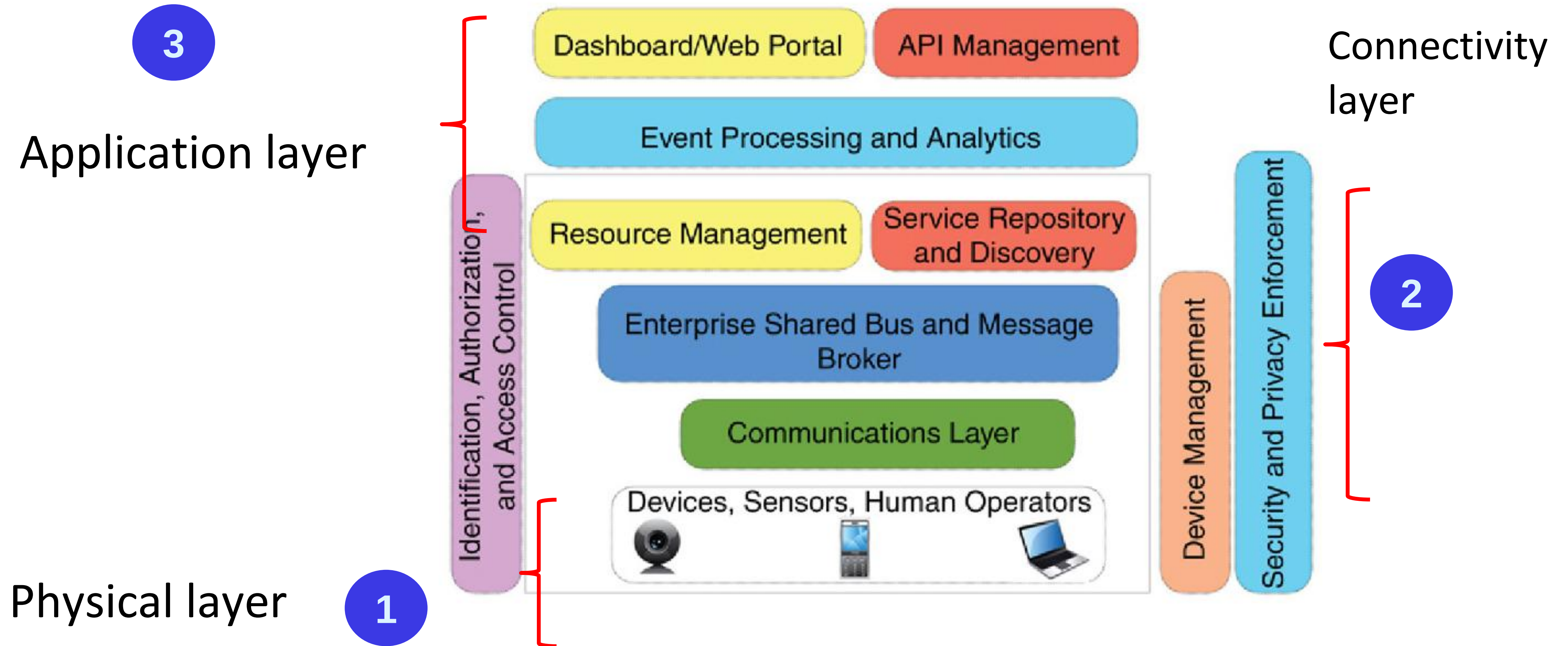
Limited devices + unreliable networks
Low power, memory, bandwidth and processing

4. LEGACY AND REAL-TIME

Old systems + immediate decisions
Gateways, edge computing and real-time analytics

Main layers (reference implementation)

11



Outline

12

Introduction

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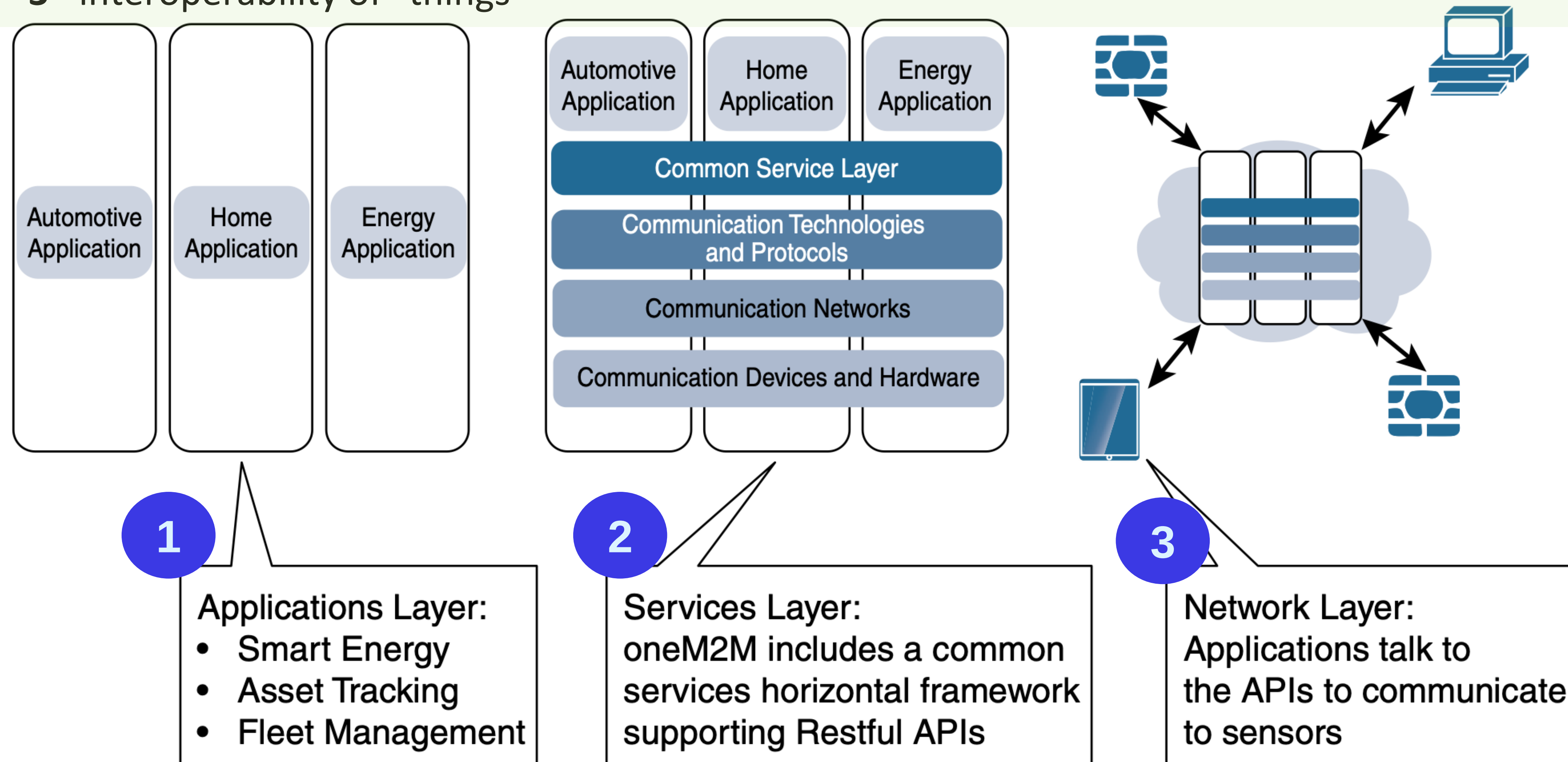
Lab: Diode

Conclusions

Initiative: oneM2M Architecture

13

- Defined by the European Telecommunications Standards Institute (ETSI)
- Created the M2M Technical Committee in 2008. Similar to the academic perspective.
- 1+2 = Core
- 3= interoperability of "things"



Initiative: IoTWF Architecture

14

- 2014: by Cisco, IBM, Rockwell Automation, and others
- 7 layers = 3 basics + others

Levels

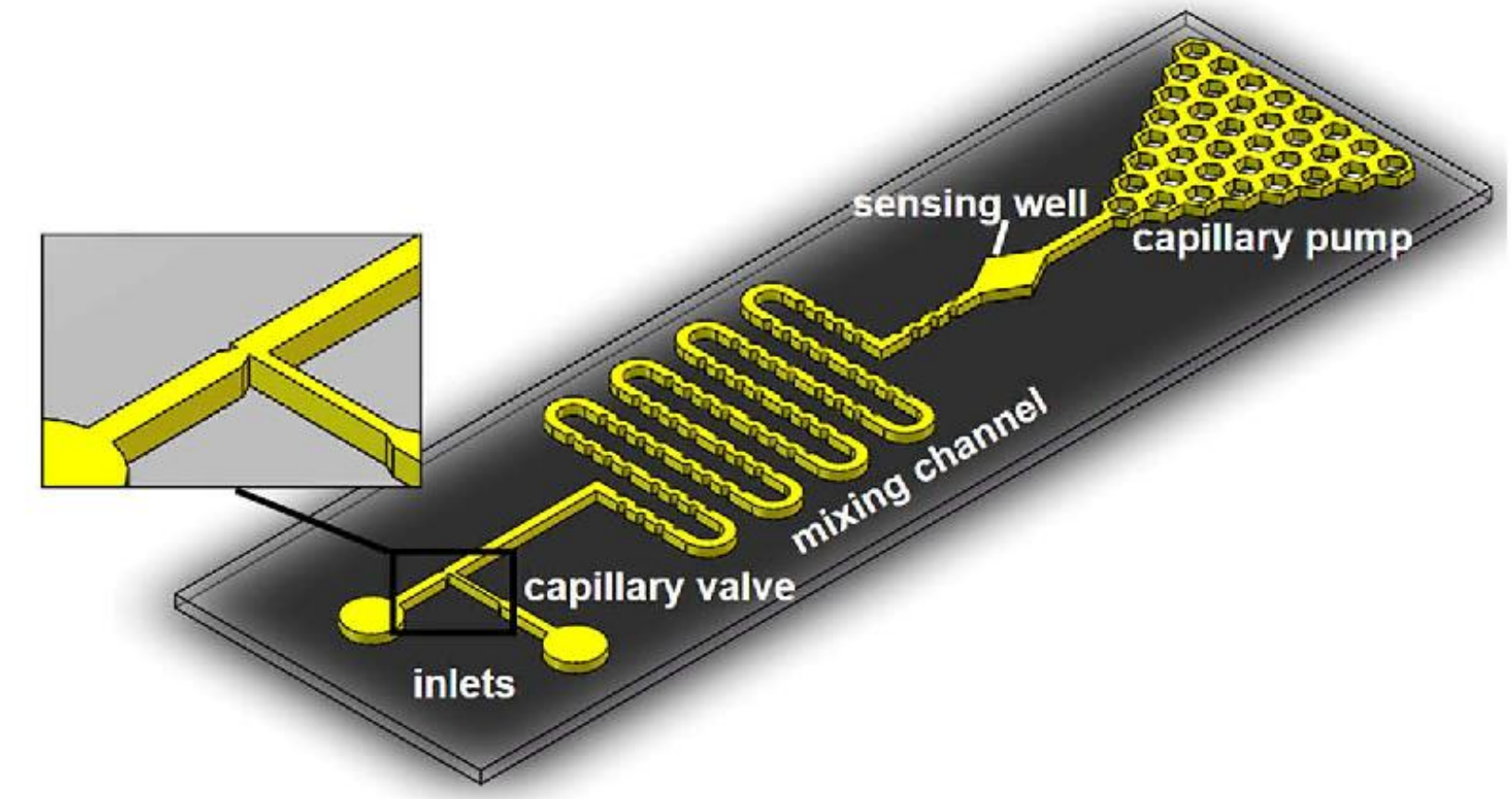
- 7 Collaboration & Processes**
(Involving People & Business Processes)
- 6 Application**
(Reporting, Analytics, Control)
- 5 Data Abstraction**
(Aggregation & Access)
- 4 Data Accumulation**
(Storage)
- 3 Edge Computing**
(Data Element Analysis & Transformation)
- 2 Connectivity**
(Communication & Processing Units)
- 1 Physical Devices & Controllers**
(The "Things" in IoT)



Layer 1

15

- The physical devices and controller's layer.
- This layer is home to the “things” on the Internet of Things, including the various endpoint devices and sensors that send and receive information.
- Their primary function is generating data and being capable of being queried and/or controlled over a network.
- The size of these “things” can range from almost microscopic sensors to giant machines.



Layer 2

16

② Connectivity (Communication and Processing Units)

Layer 2 Functions:

- Communications Between Layer 1 Devices
- Reliable Delivery of Information Across the Network
- Switching and Routing
- Translation Between Protocols
- Network Level Security



- **Guarantees the reliable and timely transmission of data.**
- **Includes transmissions between Layer 1 <->2 and 2 <->3.**
- **Doesn't distinguish between the network of the sensor/endpoint and the IoT gateway, gateway, and backhaul networks.**

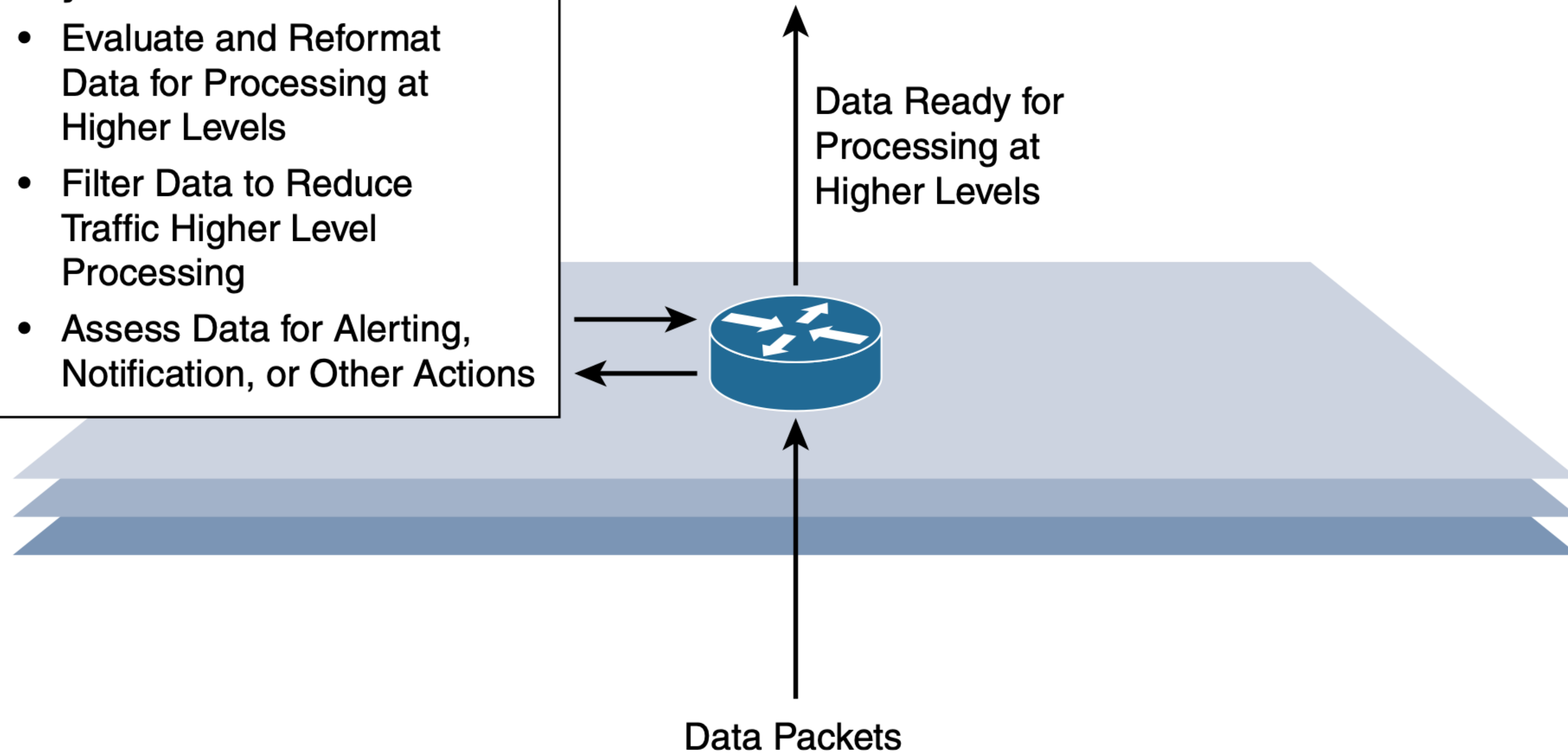
Layer 3

17

③ Edge (Fog) Computing (Data Element Analysis and Transformation)

Layer 3 Functions:

- Evaluate and Reformat Data for Processing at Higher Levels
- Filter Data to Reduce Traffic Higher Level Processing
- Assess Data for Alerting, Notification, or Other Actions



IoTWF other layers

18

IoT Reference Model Layer	Functions
Layer 4: Data accumulation layer	Captures data and stores it so it is usable by applications when necessary. Converts event-based data to query-based processing.
Layer 5: Data abstraction layer	Reconciles multiple data formats and ensures consistent semantics from various sources. Confirms that the data set is complete and consolidates data into one place or multiple data stores using virtualization.
Layer 6: Applications layer	Interprets data using software applications. Applications may monitor, control, and provide reports based on the analysis of the data.
Layer 7: Collaboration and processes layer	Consumes and shares the application information. Collaborating on and communicating IoT information often requires multiple steps, and it is what makes IoT useful. This layer can change business processes and delivers the benefits of IoT.

Commercial perspective from vendors.

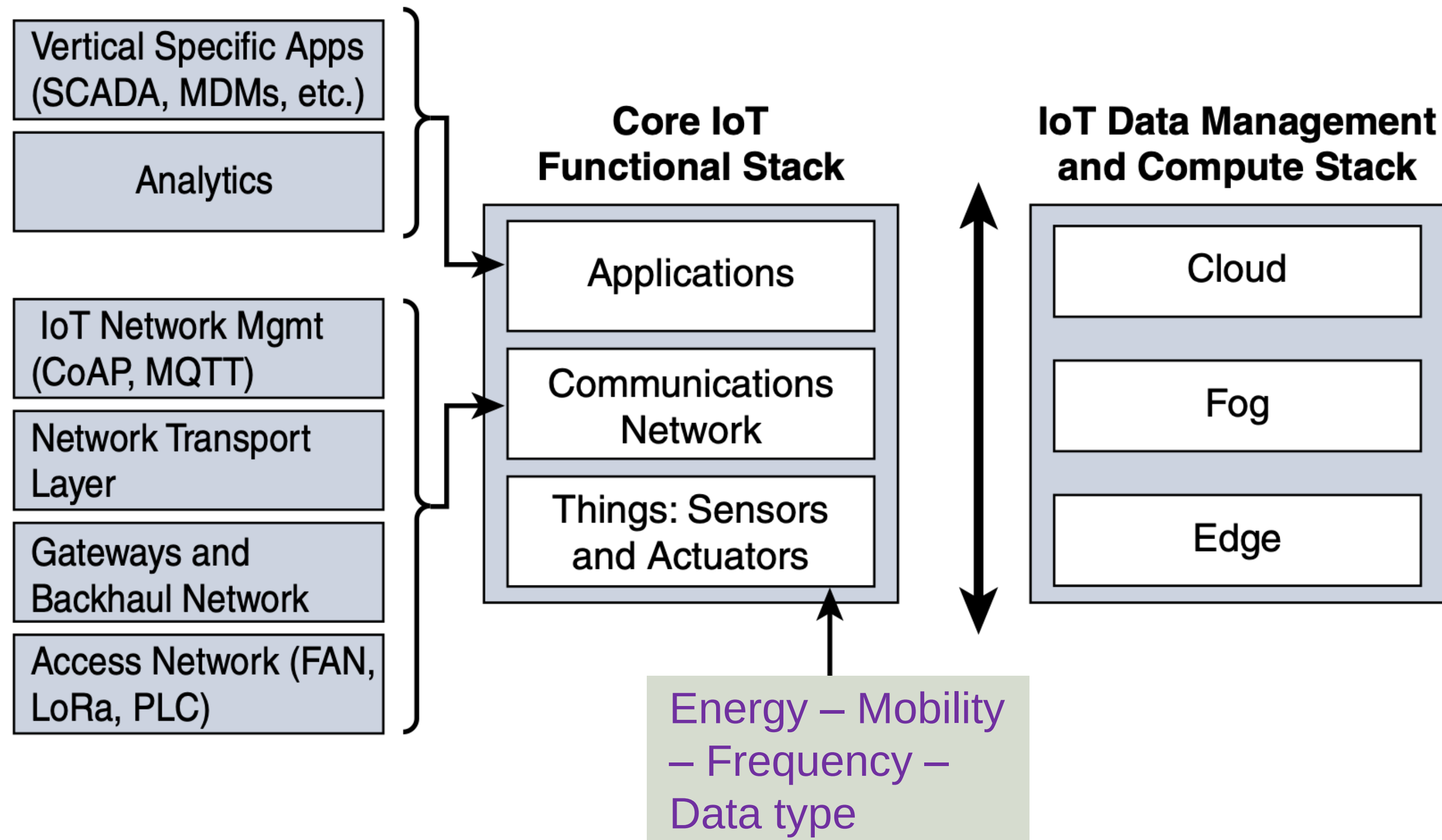
Interesting for interoperability and deployment.

Known as the IT layers

Layer 1-3 are OT layer.

General IoT Architecture

19



- **Layered architecture (abstraction to implementation)**
 - **Current IoT target** has a very important component of analytics.
- Hardware sensing platforms **are not envisioned for computing intensive tasks.**
- **Offload computing:** defines the Compute Stack.

Layer 1 – Things: Sensors & Actuators

20

The “Thing” determines the network requirements

Before choosing a network technology, we need to understand the characteristics of the device.

- **Power**

Battery-powered → energy constrained

Power-connected → fewer energy constraints

- **Mobility**

Static → fixed location

Mobile → changing location

- **Reporting Frequency**

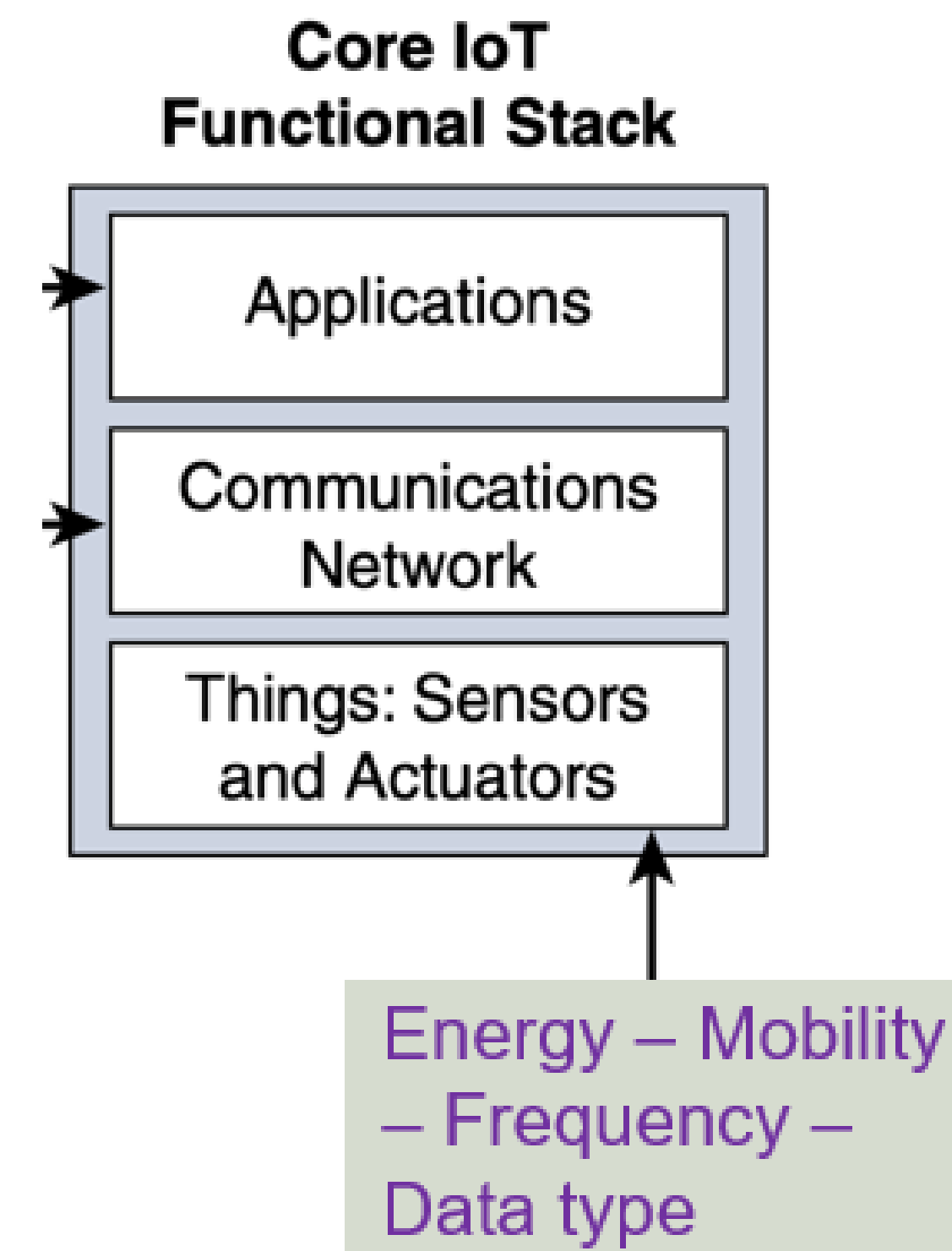
Low: once a month

High: every few milliseconds

- **Data Type**

Simple: binary / on-off

Rich: telemetry / multidimensional data / video



Layer 2 – The Communications Network

21

Connecting Things to the Applications

1. Access Network

- The “**last mile**” connecting things
- Star, mesh, or point-to-point topologies
- Examples: **Wi-SUN, LoRaWAN, IEEE 802.11ah**

2. Gateways & Backhaul

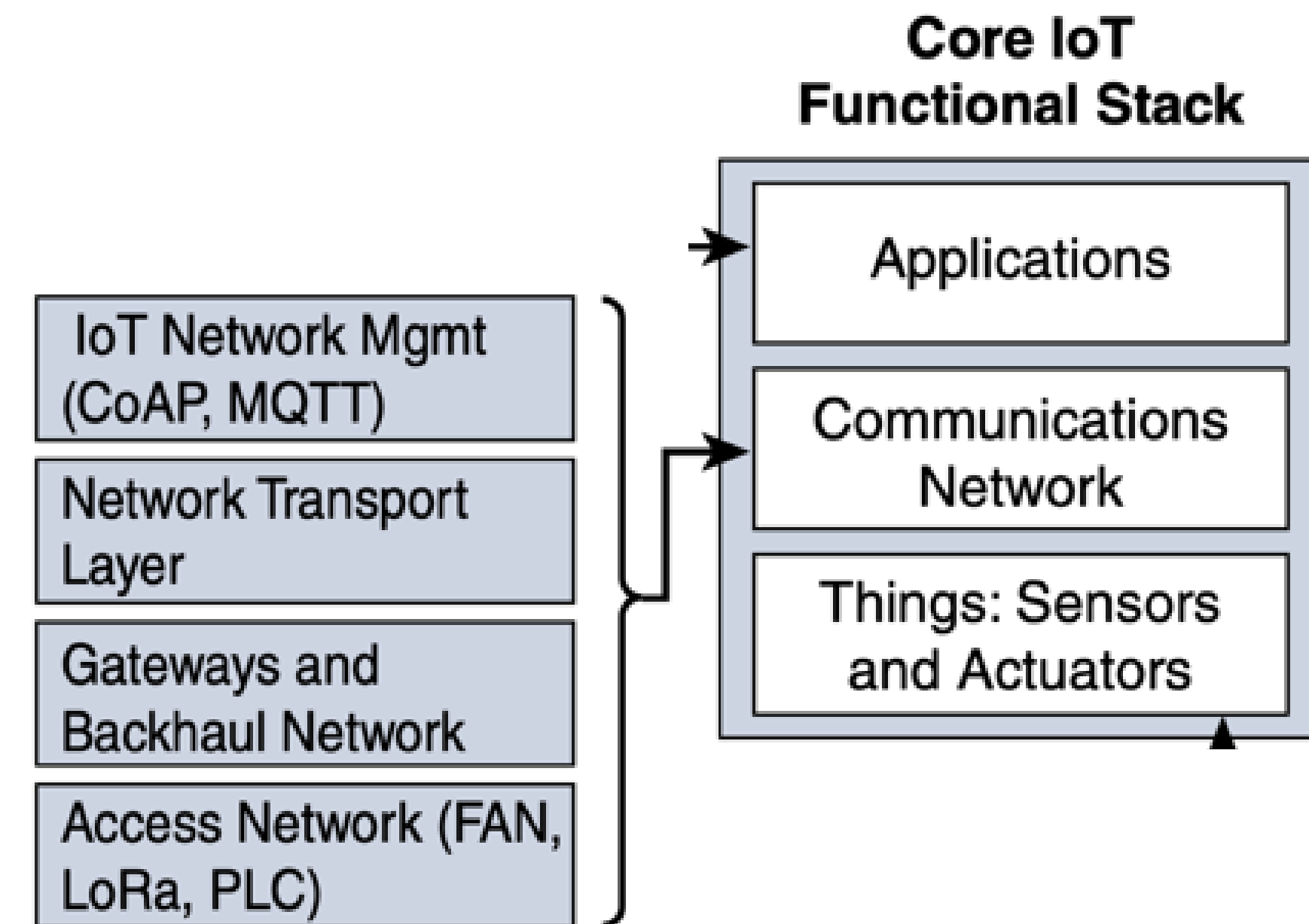
- Gateways bridge local devices and the core network
- Backhaul carries data over **longer distances**

3. Network Transport

- **IP** provides network-layer connectivity
- **UDP** enables lightweight transport

4. Network Management

- Protocols designed for efficient IoT communication
- **CoAP** → **UDP-based**
- **MQTT** → **TCP-based**



Layer 3 – Applications & Analytics

22

Turning Data into Value

Analytics Applications

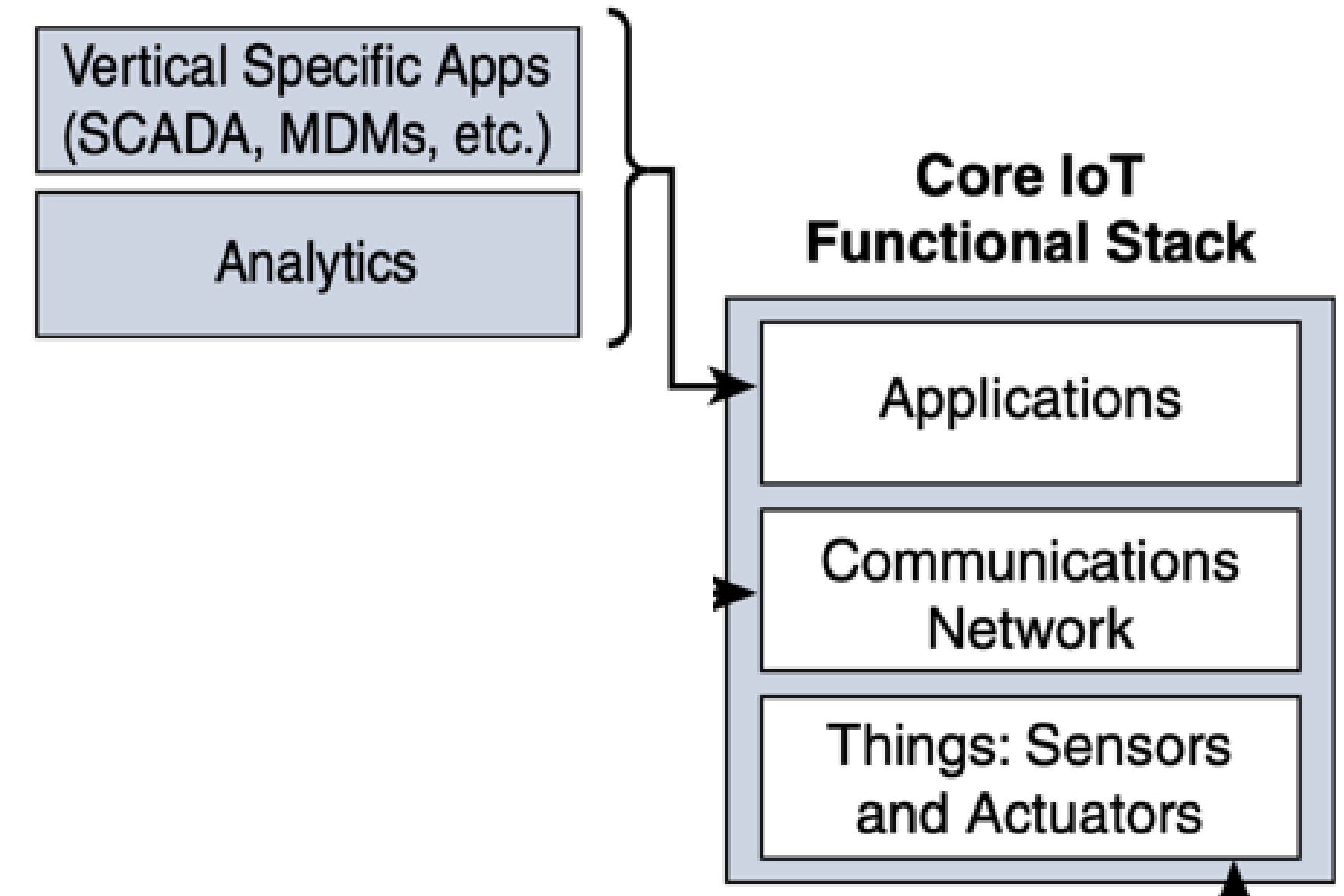
- Combine data from multiple things
- Visualize **states, trends, and statistics**
- Generate historical reports
- Identify patterns and anomalies

Control Applications

- Turn information into **automated actions**
- Modify the behaviour of physical devices

Network Analytics

- Monitor the health and performance of the IoT network
- Detect failures and abnormal behaviour
- Help prevent operational accidents



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23

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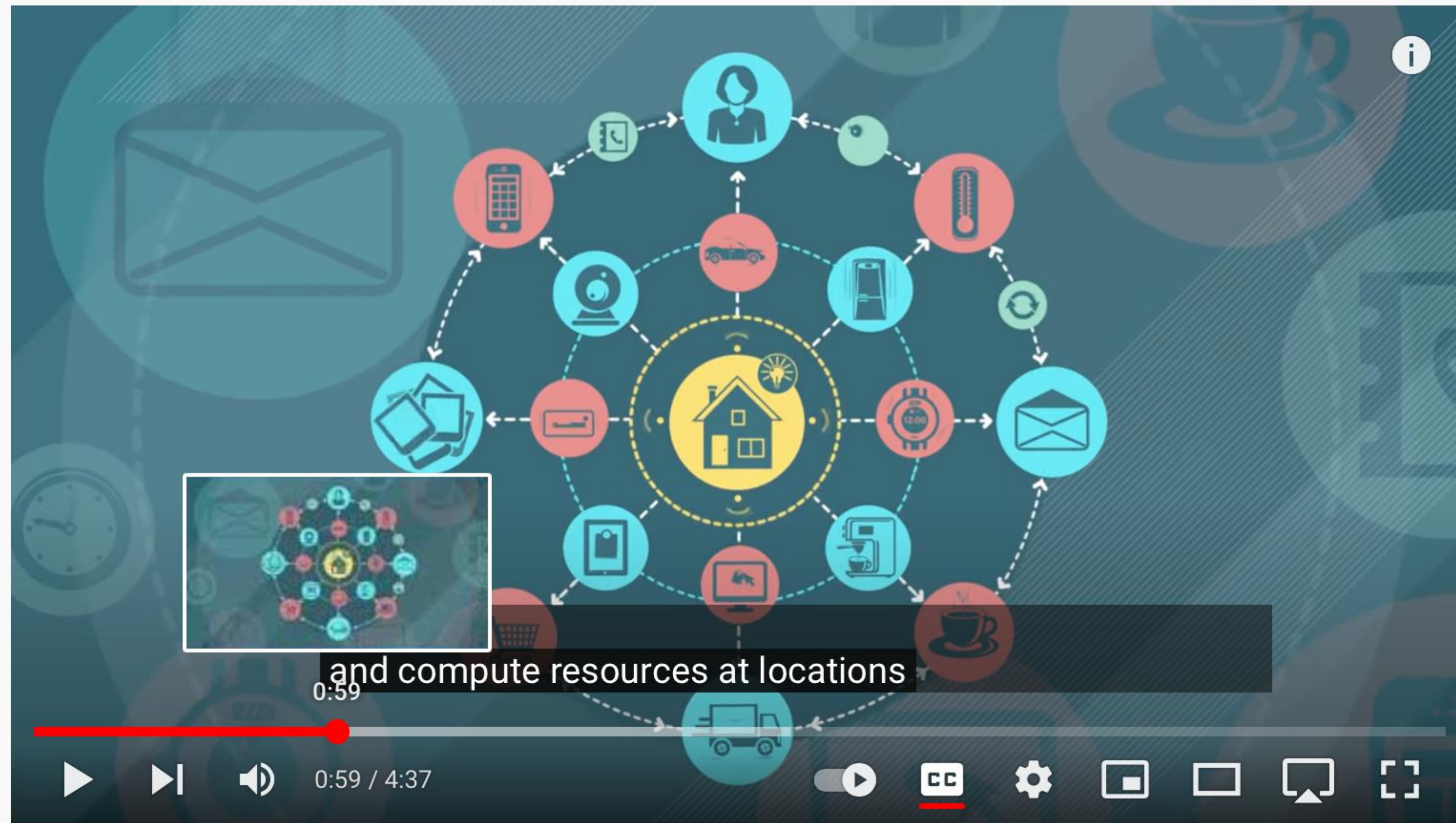
Lab: Diode

Conclusions

High level: Edge vs Cloud

24

· <https://www.youtube.com/watch?v=MBUabimpai0>

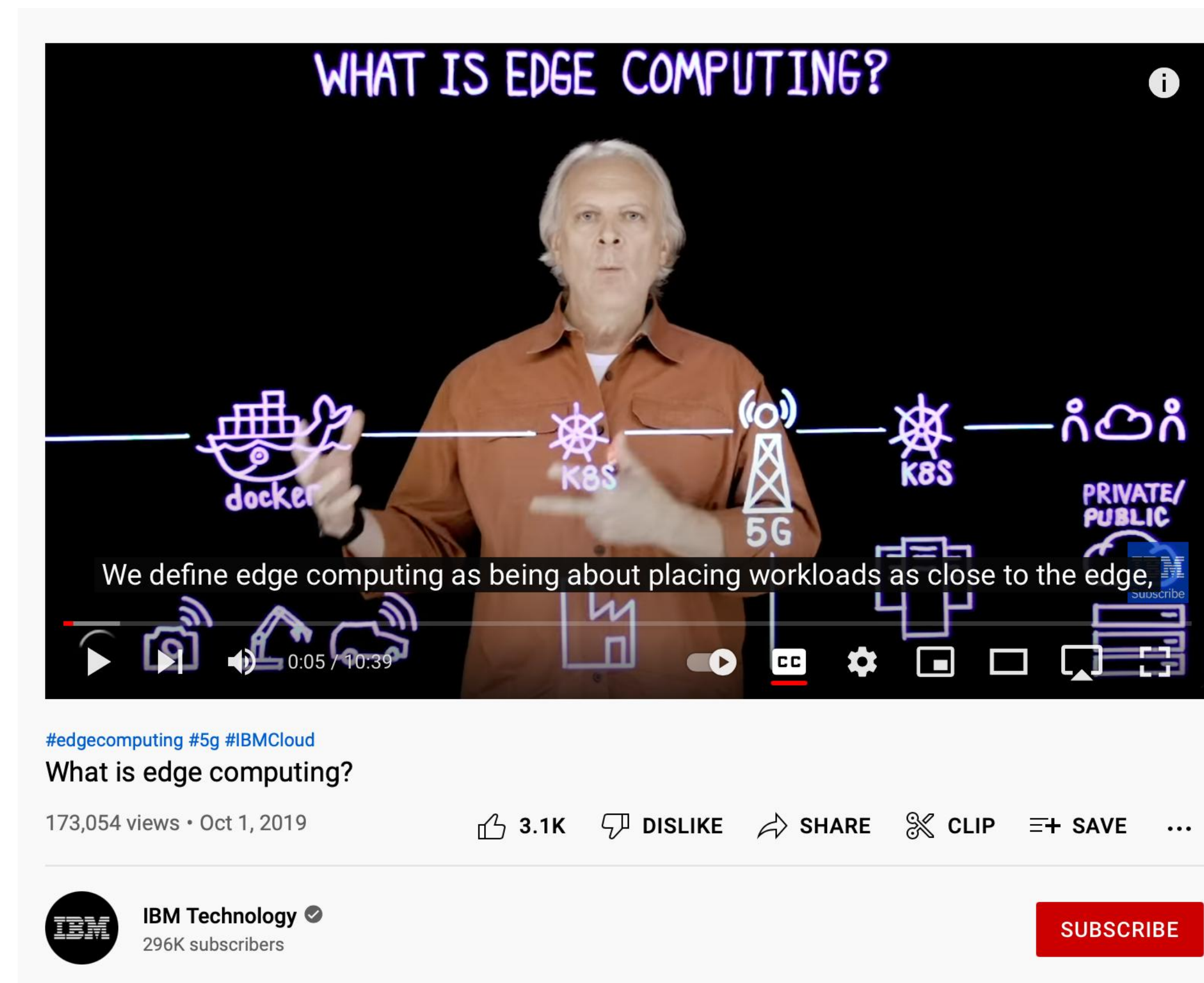


Edge Computing Vs. Cloud Computing

What is Edge Computing?

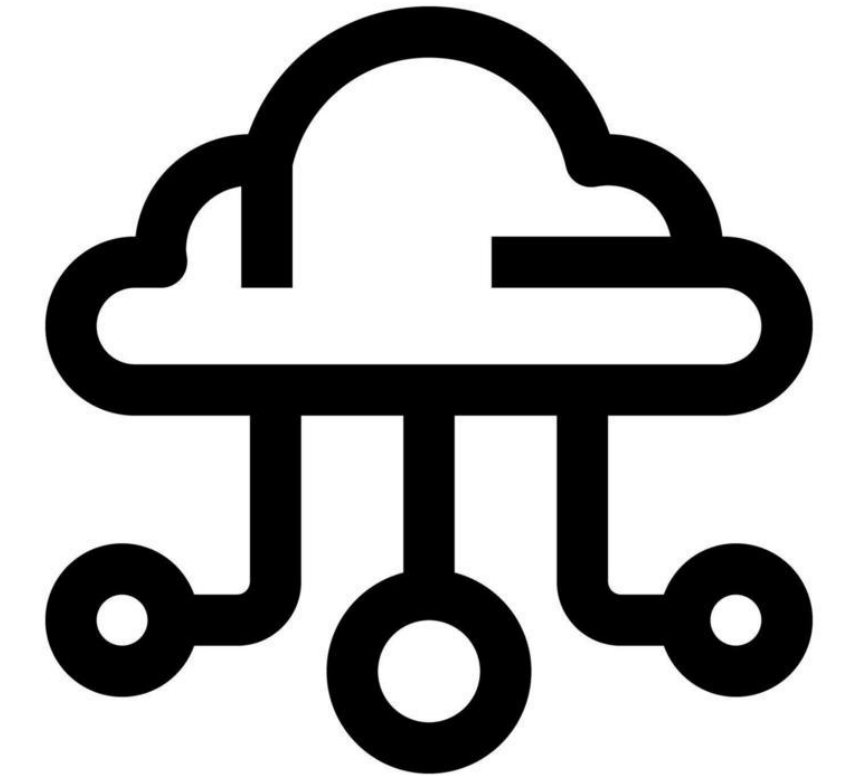
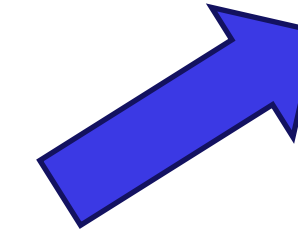
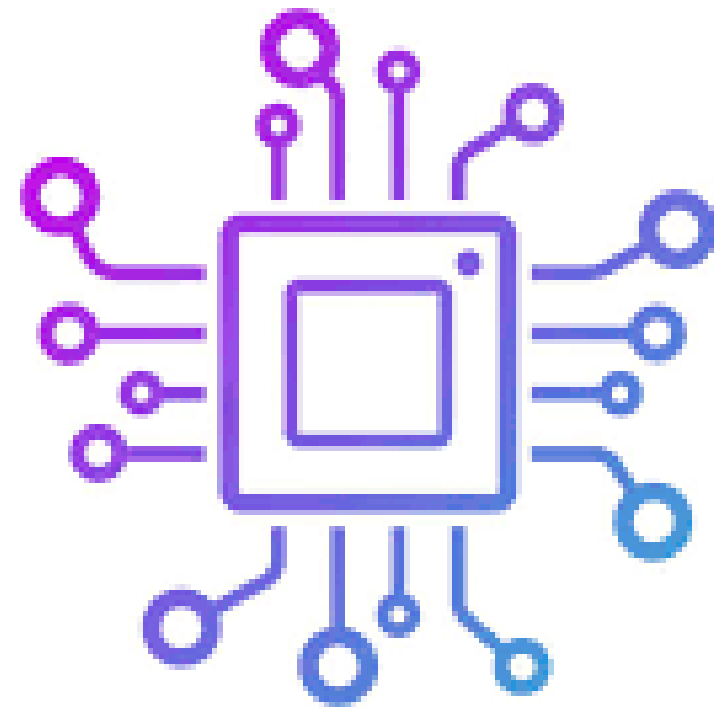
25

- <https://www.youtube.com/watch?v=cEOUeltHDdo>



IoT Gateway

26



Ex: AR502H Huawei IOT Gateway

- Protocol Translation
- Data Aggregation
- Local Processing (Edge/Fog Computing)
- Connectivity Management
- Security Functions

Cloud vs Fog (Edge)

27

	Fog	Cloud
Response time	Low	High
Availability	Low	High
Security level	Medium to hard	Easy to medium
Service focus	Edge devices	Network/enterprise core services
Cost for each device	Low	High
Dominant architecture	Distributed	Central/distributed
Main content generator—consumer	Smart devices—humans and devices	Humans—end devices

- It is important to notice that both execute on distributed systems.
- In this case the idea of centralized system can refer to physical location.

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28

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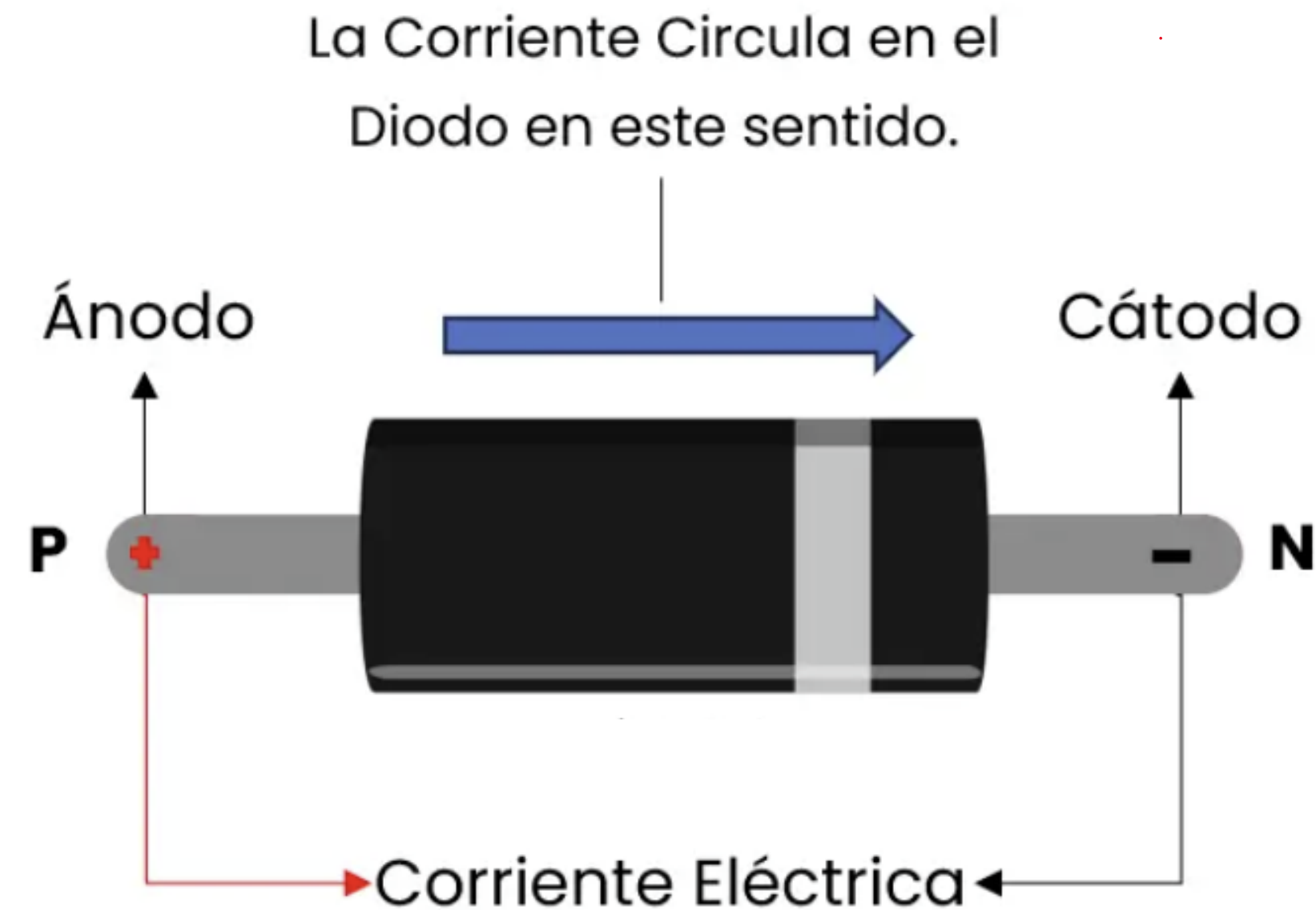
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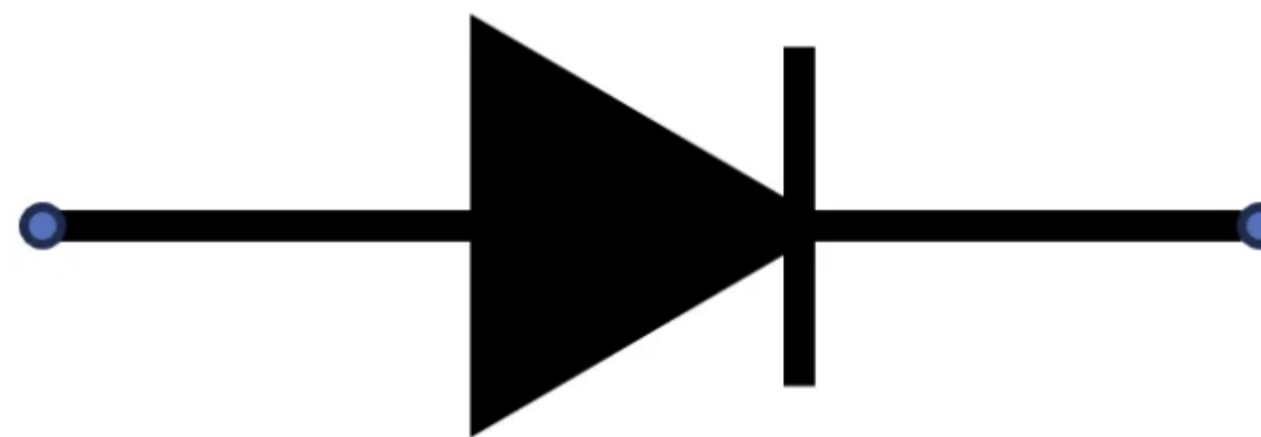
Diode

29

- El terminal positivo de un diodo (cuando está en polarización directa).
- En el símbolo del diodo, el ánodo es el extremo sin la barra.



- Es el terminal negativo de un diodo (cuando está en polarización directa).
- Es el lado por donde sale la corriente convencional.
- En el símbolo, es el lado con la barra ($| \blacktriangleright$).

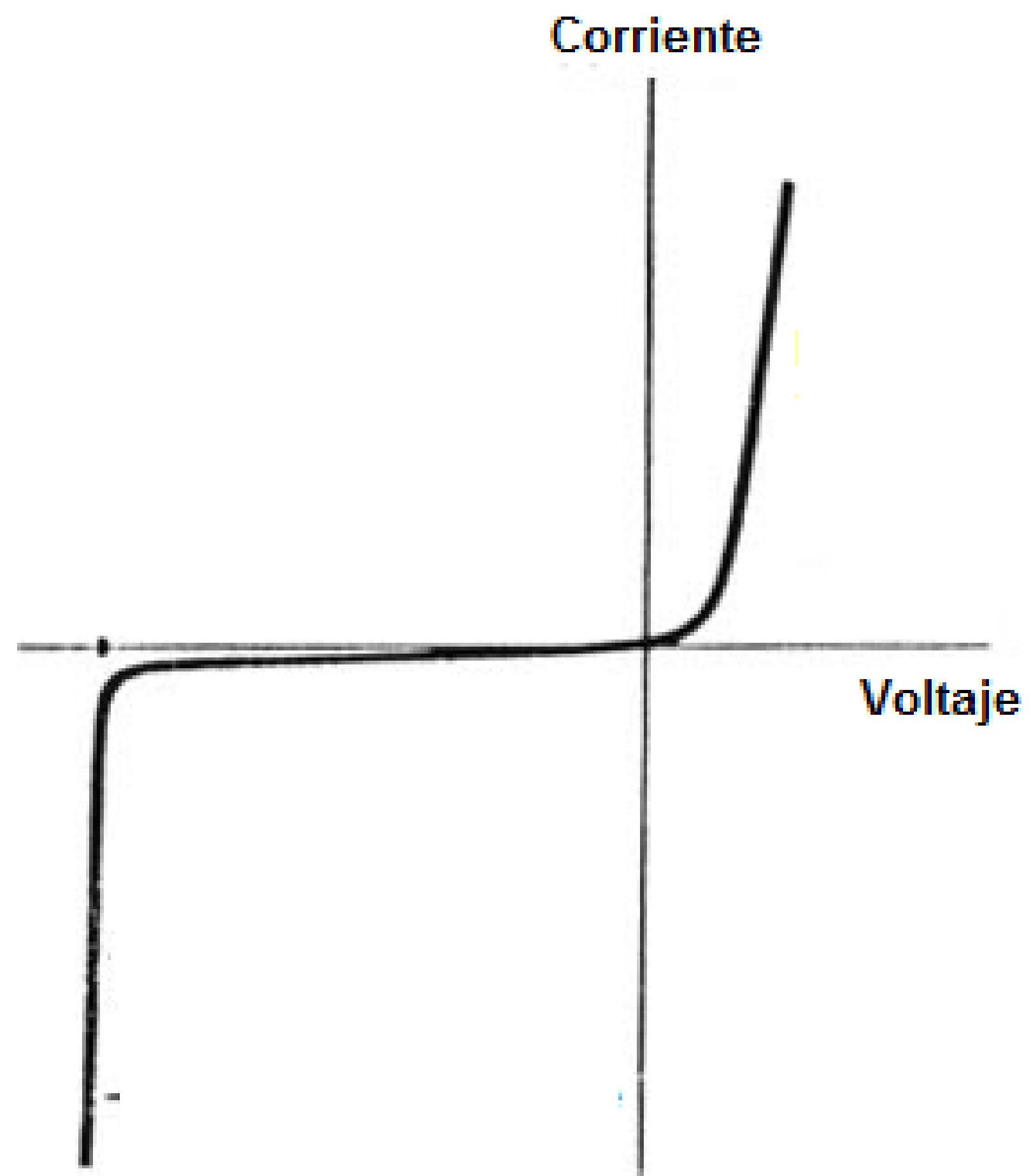
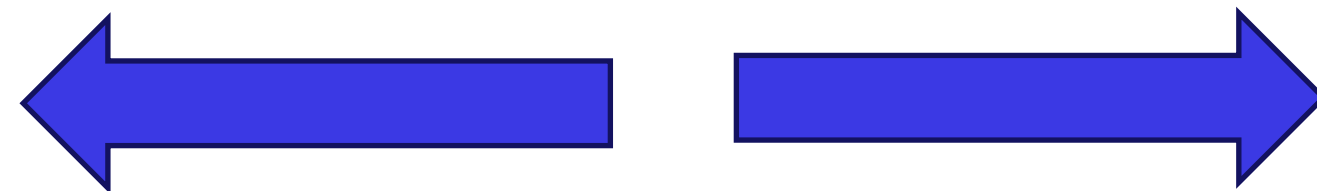


Diode

30

Polarización inversa

Polarización directa



Voltaje umbral: V_f

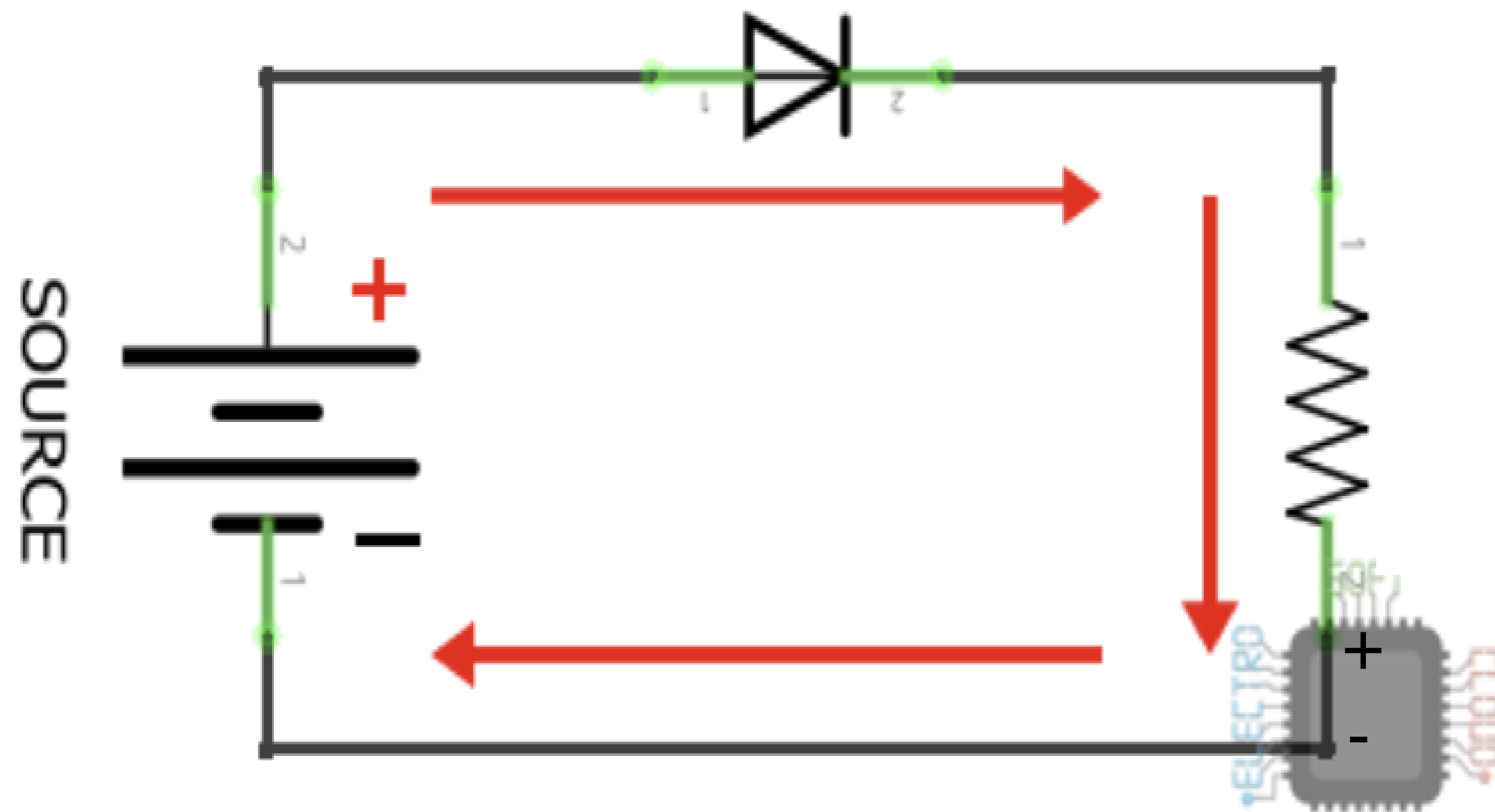
Corriente de fuga inversa: I_r

Voltaje de ruptura: V_{br}

Corriente directa I_f

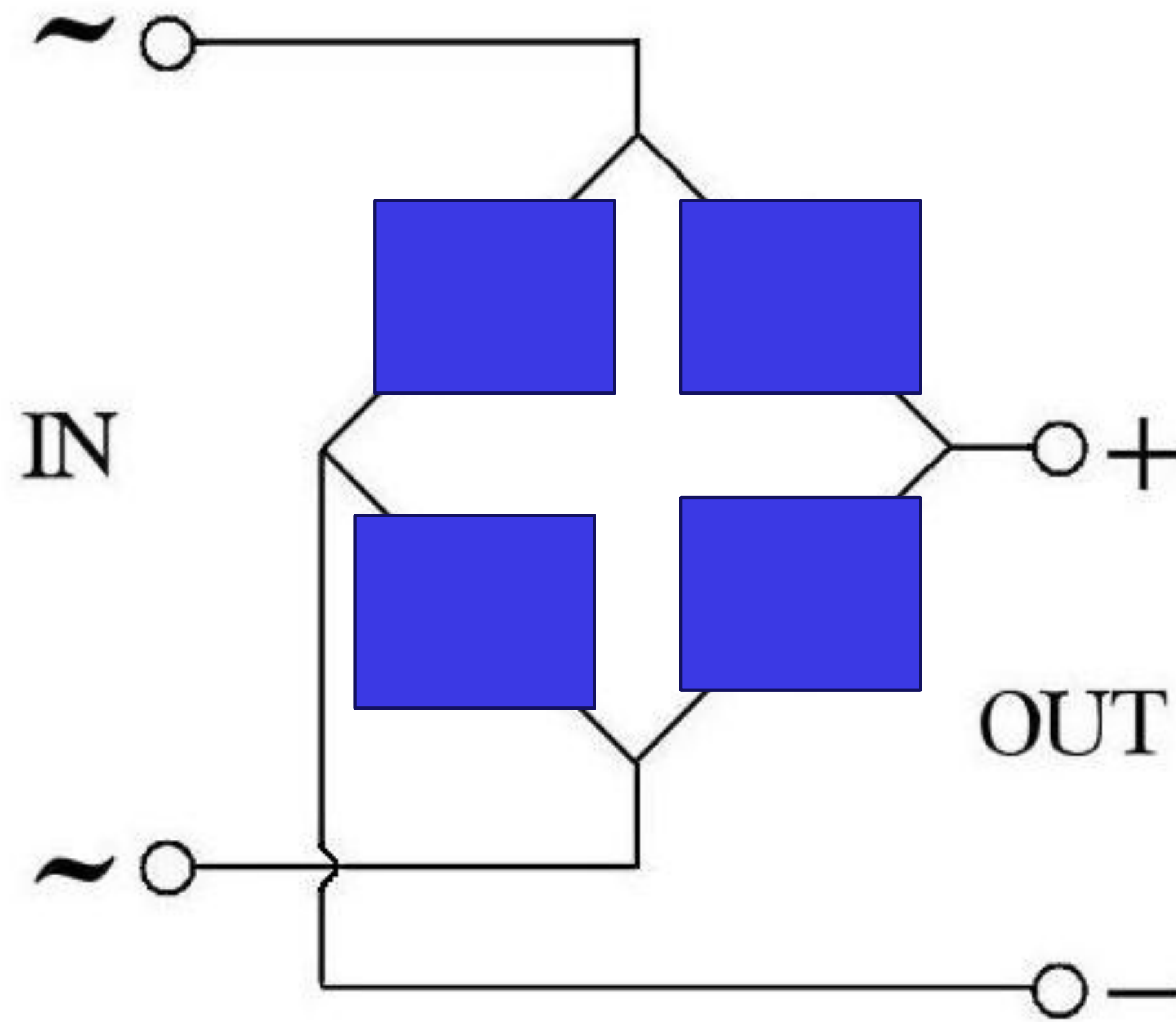
Question1 Diode protection

31



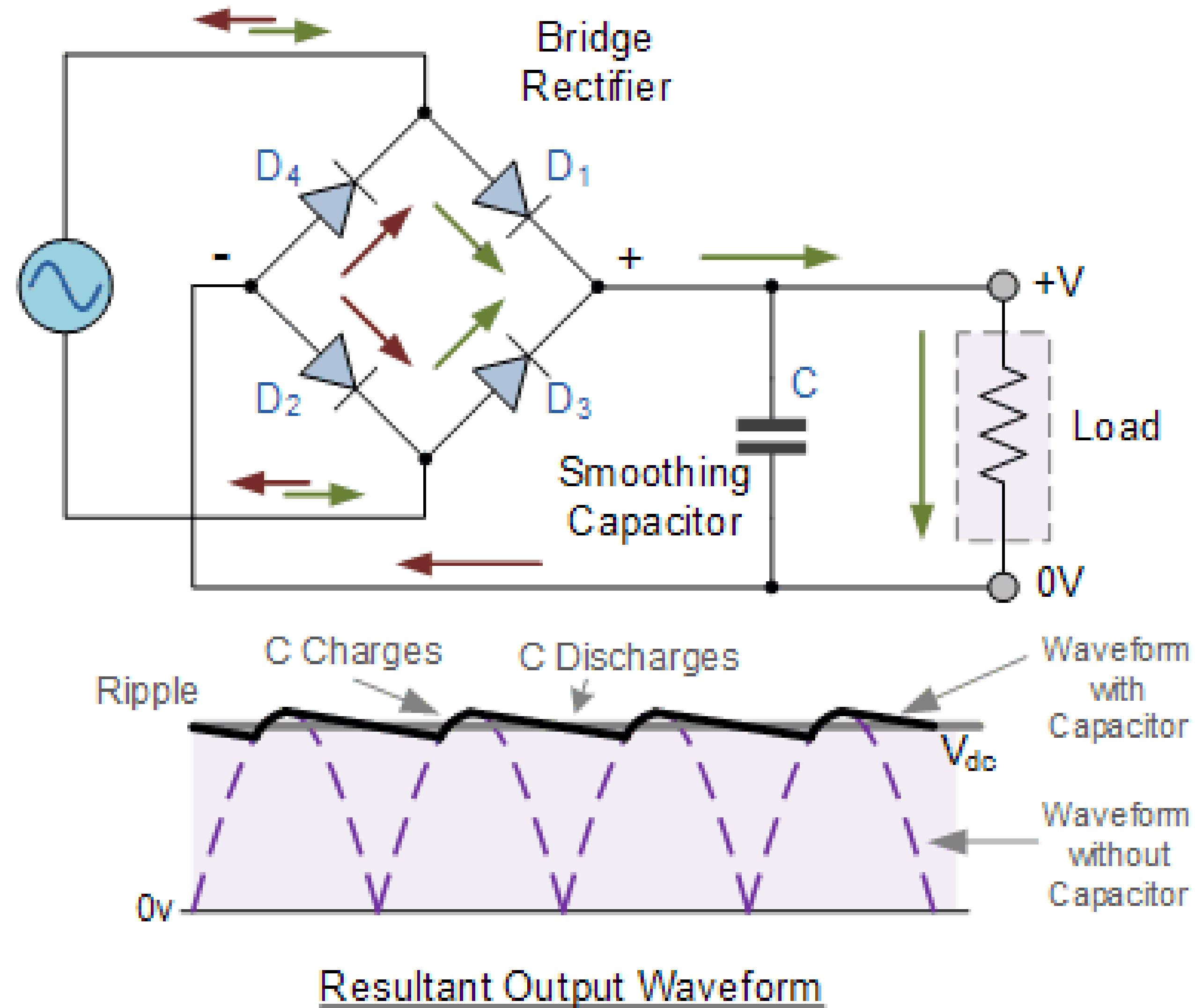
Question 2- Rectifier

32



Full wave rectifier circuit

33



Outline

34

Introduction

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Lab: Diode

Conclusions

- IoT architecture types and layered models were discussed.
- Examples of architectures were reviewed, including oneM2M and IoTWF.
- Edge, fog, and cloud computing concepts were introduced, along with their trade-offs.
- The diode concept and some of its applications were introduced.

2. IoT Architecture

Internet of Things